

Congruence & Similarity

Test · 36 marks total · Section D is Higher-tier content

Name: _____

Date: _____

Score: _____ / 36

Section A — Congruence

A1. Explain the difference between two shapes being *congruent* and two shapes being *similar*.

[2 marks]

A2. For each pair of triangles, state the condition (SSS, SAS, ASA or RHS) that proves they are congruent.

(a) Both have sides 6 cm, 9 cm and 11 cm.

(b) Both are right-angled, with hypotenuse 10 cm and one other side 6 cm.

(c) Both have sides 7 cm and 5 cm with an angle of 55° between them.

[3 marks]

A3. Two triangles both have angles of 40° , 60° and 80° . Explain why this does not prove they are congruent.

[1 mark]

A4. ABCD is a parallelogram with diagonal AC. Prove that triangles ABC and CDA are congruent.

[3 marks]

Section B — Similar shapes: missing lengths

B1. A rectangle measures 4 cm by 6 cm. A similar rectangle has a shorter side of 10 cm. Find its longer side.

[2 marks]

B2. Triangles ABC and PQR are similar, with A matching P, B matching Q and C matching R. $AB = 5$ cm, $BC = 8$ cm and $AC = 7$ cm. $PQ = 12.5$ cm. Find QR and PR.

[3 marks]

B3. Rectangle X is 3 cm by 5 cm. Rectangle Y is 6 cm by 9 cm. Are the rectangles similar? Show how you decide.

[2 marks]

B4. Two triangles are similar. A side of 18 cm on the larger triangle matches a side of 12 cm on the smaller one. Another side of the larger triangle is 24 cm. Find the matching side on the smaller triangle.

[2 marks]

Section C — Similar triangles inside each other

In triangle ABC, D lies on AB and E lies on AC, with DE parallel to BC.

C1. Explain why triangle ADE is similar to triangle ABC.

[1 mark]

C2. AD = 4 cm, DB = 6 cm and DE = 5 cm. Find the length of BC.

[3 marks]

C3. In a different triangle of the same type, AE = 6 cm, AC = 15 cm and BC = 20 cm. Find the length of DE.

[2 marks]

Section D — Area & volume of similar shapes (Higher)

D1. A shape is enlarged by length scale factor 3. By what scale factor is its area multiplied?

[1 mark]

D2. Two similar triangles have matching sides of 4 cm and 12 cm. The smaller triangle has area 10 cm^2 . Find the area of the larger triangle.

[2 marks]

D3. Two similar cones have heights 6 cm and 9 cm. The smaller cone has volume 64 cm^3 . Find the volume of the larger cone.

[3 marks]

D4. Two similar solids have surface areas 50 cm^2 and 200 cm^2 . The smaller solid has volume 30 cm^3 . Find the volume of the larger solid.

[4 marks]

D5. The volumes of two similar jugs are in the ratio 8 : 27. Find the ratio of their heights.

[2 marks]